

Directories & Files

Unix-like systems store data and programs in files. These are organized in directories. Directories are data constructs (metadata) in an index node (or inode). The directory inode entry contains a list of inode entries for files and other directories.

A directory inode contains a lot of information, including the following fields:

Table 5.5: List of Disk Formatting Tools and Their Functionalities

FIELD NAME	DESCRIPTION
inode	Unique number identifying the file object
name_len	Length of the file name
file_type	0=unknown, 1=regular-file, 2=directory, 3=character-device, 4=block-device, 5=FIFO, 6=socket, 7=symbolic-link
name	File name

To reach the information for a file, first, the inode of the directory is read, and then, the selected file inode number is read. The file inode contains data elements associated with the file. Some of the file-related information stored in the file inode are (you can use the command `stat filename`):

- The size of the file in bytes.
- User ID of the file.
- Group ID of the file.
- The file mode that determines the file type and how the owner, group, and others (world) can access the file.
- Additional system and user flags to further protect the file.
- Timestamps telling when the inode itself was last changed (`ctime` or changing time), the file content was last modified/written (`mtime` or modification time), and when the file was last accessed/read (`atime` or access time).
- A link counter that lists how many hard links point to the inode.
- Pointers to the disk blocks that store the file's content.

The Filesystem Hierarchy Standard (FHS) defines a suggested logical location of directories and files on Linux distributions. These definitions include what should be contained in specific directories and files.

Below, you can see a screenshot of the directory structure in Ubuntu:



```
student@ubuntu:~$ tree -L 1 -d /
/
├── bin -> usr/bin
├── boot
├── cdrom
├── dev
├── etc
├── home
├── lib -> usr/lib
├── lib32 -> usr/lib32
├── lib64 -> usr/lib64
├── libx32 -> usr/libx32
├── lost+found
├── media
├── mnt
├── opt
├── proc
├── root
├── run
├── sbin -> usr/sbin
├── snap
├── srv
├── sys
├── tmp
├── usr
└── var

24 directories
student@ubuntu:~$
```

Directory Structure

These are the files that are required in the root directory ("/"). The root directory is the highest level or the top level directory for Linux systems.

Man pages are reference software documentation found on Unix operating systems, including Linux. You can learn more about the Linux filesystem hierarchy from such a man page. This page can be accessed locally on your workstation using the following command:

```
$ man hier
```

Man pages are also available online; for example, the page mentioned above can be accessed online as well: [man hier](#).